

Skip-Vision: Efficient and Scalable Acceleration of Vision-Language Models via Adaptive Token Skipping

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Abstract

Transformer-based models have driven significant advancements in Multimodal Large Language Models (MLLMs), yet their computational costs surge drastically when scaling resolution, training data, and model parameters. A key bottleneck stems from the proliferation of visual tokens required for fine-grained image understanding. We propose Skip-Vision, a unified framework addressing both training and inference inefficiencies in vision-language models. On top of conventional token compression approaches, our method introduces two complementary acceleration strategies. For training acceleration, we observe that Feed-Forward Network (FFN) computations on visual tokens induce marginal feature updates. This motivates our Skip-FFN strategy, which bypasses FFN layers for redundant visual tokens. For inference acceleration, we design a selective KV-cache removal mechanism that prunes the skipped key-value pairs during decoding while preserving model performance. Experimental results demonstrate that Skip-Vision reduces training time by up to 35%, inference FLOPs by 75%, and latency by 45%, while achieving comparable or superior performance to existing methods. Our work provides a practical solution for scaling high-performance MLLMs with enhanced efficiency. Our project page can be found [here](#).

1. Introduction

Transformer-based large-scale models have significantly advanced progress in Artificial Intelligence (AI) [10, 86, 101]. Their remarkable capabilities have driven the emergence of modern Multimodal Large Language Models (MLLMs) [2, 3, 69, 88], which demonstrate vision-language competencies comparable to human performance. Empirical evidence suggests that scaling laws continue to

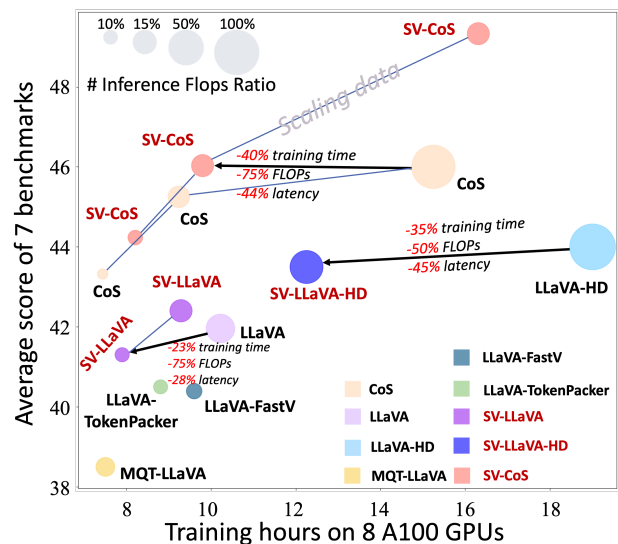


Figure 1. **Performance-efficiency trade-off curve.** Each circle denotes a model configuration, where our models utilize the Skip-Vision framework, with CoS [40], LLaVA-HD [67] and LLaVA [69] serving as baselines. Circle sizes reflect the inference FLOPs ratio. Skip-Vision demonstrate superior performance, scaling effectively with increased FLOPs and data and achieving higher inference efficiency when compared to baselines and other efficient MLLM methods. All methods utilize LLaMA3 8B as the foundational large language model.

be effective, primarily across three dimensions: visual scaling [40, 67, 112], data scaling [63, 64], and model scaling [56, 104]. However, these scaling methods present substantial efficiency issues, significantly increasing the training and inference burden. In a typical LLaVA-1.5 framework [69], where the model receives 576 visual tokens and 40 text tokens, inference using Vicuna-13B [25] LLM backbone demands 200 trillion FLOPs [42]. These motivate our pursuit of an MLLM architecture that optimally balances efficiency and performance.

Recent advances in Multimodal Large Language Models (MLLMs), such as Chain-of-Sight (CoS)[40] and LLaVA-

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Llama layer	Complexity	Parameters
FFN	$10.5LNC^2$	$5.64B(70\%)$
Attention	$2LN^2C + 1.5NC^2$	$1.34B(17\%)$
Embedding	-	$0.5B(6\%)$

Table 1. **Computational complexity and parameter statistics for Llama3 8B.** L denotes the number of blocks, N represents the number of input tokens, and C is the feature dimensionality

Next[68], have improved performance by increasing visual tokens. However, during the Supervised Fine-Tuning (SFT) stage, the computational cost rises dramatically, leading to diminishing returns and making the scaling unsustainable. As shown in Figure 1, CoS requires a 165% increase in training resources and a 400% increase in inference computation for a mere 1.7% performance improvement, while LLaVA-Next faces similar inefficiencies. This challenge emphasizes the need to reduce visual token computation. Approaches like token compression or pruning [15, 22, 39, 60, 116, 119], while efficient, risk losing critical visual information, revealing a trade-off between efficiency and performance. This work aims to optimize the model’s computational efficiency from an architectural perspective, proposing a scalable solution that balances efficiency with visual fidelity.

We systematically examine the parameter count and FLOPs distribution across different components of the LLM. As shown in Table 1, using LLaMA-3 [29] as an example, the parameter and FLOPS requirements for the FFN are significantly higher than the attention module. Furthermore, as illustrated in Figure 2, when the number of visual tokens is large, the majority of computations are concentrated on FFN of these tokens. In particular, we computed the difference in the magnitudes of the features before and after the FFN transformation for different tokens. As illustrated in the figure 3, we observe that the updates introduced by the FFN to the visual tokens are significantly smaller than those that affect the text tokens. This is similar to the observation in [123]. This observation suggests a potential approach: **could we bypass FFN computations for a substantial portion of visual tokens?** Furthermore, in the autoregressive training framework of MLLMs, only the final visual token utilizes its last hidden state for next-token prediction, while the other visual tokens primarily serve to convey causal information. During model inference, inspired by [7], we observe that most visual information is progressively integrated into the final token through the early causal layers. This raises the possibility of reducing reliance on the full visual scaling KV cache throughout inference, potentially accelerating the inference process.

Building on these insights, we propose Skip-Vision, a novel and flexible architecture designed for more efficient scaling (see Figure 1 for a comparison). Visual tokens de-

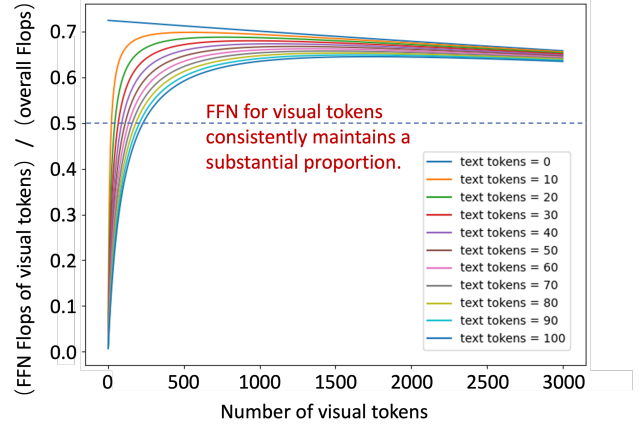


Figure 2. **Flops ratio of visual tokens in FFN.** When visual tokens greatly outnumber text tokens, computation is predominantly consumed by the FFN of visual tokens. Reducing their passage through the FFN can thus markedly decrease computational costs.

rived from smaller window/scale sizes constitute approximately 80% of the total visual tokens and account for most of the computational overhead. To mitigate this, as shown in Figure 4, we allow these tokens to skip the Feed-Forward Network (FFN) computation, significantly reducing overall computational costs. To address redundancy within skipped vision tokens, we can apply a token merging strategy for further simplification. During inference, skipped tokens are removed from the KV cache after the pre-fill phase, accelerating subsequent computation. Empirical results demonstrate that Skip-Vision reduces training time by up to 35%, FLOPs of inference by 75% and inference latency by 45%, while maintaining or even improving performance. These efficiency gains provide a scalable solution for large-scale multimodal learning, enabling better data utilization and enhanced model scalability.

In summary, Skip-Vision is designed to be seamlessly integrated into the standard SFT pipeline of MLLMs without introducing additional retraining or decoupled modules. It directly modifies the transformer’s computation flow, offering a practical and theoretically grounded acceleration solution for MLLM training and inference jointly. Our contributions are as follows:

- We introduce a novel and efficient framework, using token merge and skip FFN strategy during training to reduce redundant computation for visual tokens.
- In inference, our framework employs a skip KV-cache mechanism that removes skip-FFN visual tokens from the KV-cache, enhancing efficiency.
- We present a theoretical analysis of the performance error in Skip-Vision and establish an error bound, which will be empirically validated through experiments.
- Experiments show our model’s superior efficiency, effective data scaling, and performance on par with state-of-

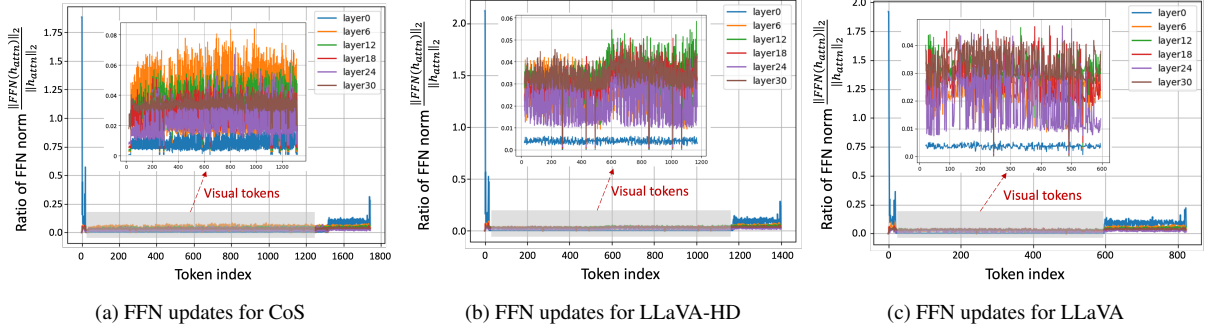


Figure 3. **FFN impact for different MLLMs.** We evaluate FFN impact by computing the feature modulus ratio before ($\|h_{attn}\|_2$) and after FFN ($\|FFN(h_{attn})\|_2$). The FFN introduces significantly smaller updates to visual tokens compared to text tokens.

the-art models of similar scale.

2. Related work

2.1. Vision-Language models

Rapid progress in large language models (LLMs) [25, 29, 114] has laid a solid foundation for the emergence of large-scale vision-language models (VLMs) [3, 6, 19, 58]. Flamingo [3] was a pioneering effort, integrating pre-trained image encoders with LLMs to handle multimodal data, thereby allowing comprehensive understanding and reasoning across modalities. Following this, the major models developed by leading corporations, such as GPT-4v [2] and Gemini-1.5-Pro [88], have led the progress in this domain. Using proprietary data sets and undisclosed training methodologies, these models have elevated multimodal intelligence to unprecedented levels.

At the same time, the open-source community has worked diligently to stay abreast of these advancements. The LLaVA series models [56, 67, 69, 112] exemplify a key strategy in this domain by mapping visual features directly into the input embedding space of the language model, integrating them as input tokens. However, this approach generates a substantial volume of visual tokens, frequently exceeding the number of text tokens, creating efficiency challenges. InternVL-1.5 [24] adopts dynamic high-resolution techniques, segmenting large images into smaller fragments for processing and scaling them accordingly. MiniCPM-V [117] utilizes a perceiver resampler structure with a single layer of cross-attention to compress visual tokens. Models such as Vary [107], SPHINX [64], Cambrian-1 [100], and BRAVE [47] leverage multi-visual encoder architectures to strengthen their visual processing capabilities. Furthermore, a crucial factor contributing to the effectiveness of modern VLMs is instruction fine-tuning [28, 67, 134], enabling these models to function as conversational agents and interact with users through natural, human-like dialogue.

2.2. Efficient multimodal large language models

To enhance the efficiency of vision-language models, one of the primary and widely employed approaches in contemporary research involves reducing the size of the backbone models [27, 124, 133]. Simultaneously, given that visual tokens contribute significantly to computational demands, research focused on compressing or refining these tokens has garnered increasing attention. Such methods are often coupled with the design of vision-language projectors. For instance, techniques like the Perceiver Resampler [3, 6, 41] and Q-Former [58] utilize transformer-based mechanisms to consolidate visual tokens into more compact query sets. In this way, the transformer outputs corresponding to the positions of the learnable latent queries serve as the aggregated representation of the visual features [42]. Honeybee [14] propose two visual projectors, namely C-Abstractor and D-Abstractor. LLaVA-PruMerge [91] and MADTP [12] introduce adaptive methods for reducing visual tokens. In addition, while the large language model (LLM) field has employed various techniques for token reduction to accelerate inference and compress KV cache [35, 131], research in this area remains limited within the MLLM domain.

3. Preliminaries

3.1. Decoder-only LLM

Each Transformer layer in a decoder-only LLM primarily consists of a self-attention layer and a feed-forward neural network. To facilitate autoregressive inference, the use of a KV-cache has been developed to accelerate the decoding process.

Self-Attention Mechanism: The self-attention mechanism enables each token in a sequence to selectively attend to other tokens, capturing contextual dependencies. Each token generates a query (Q), key (K), and value (V) vector. Attention scores are calculated by the scaled dot-product:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (1)$$

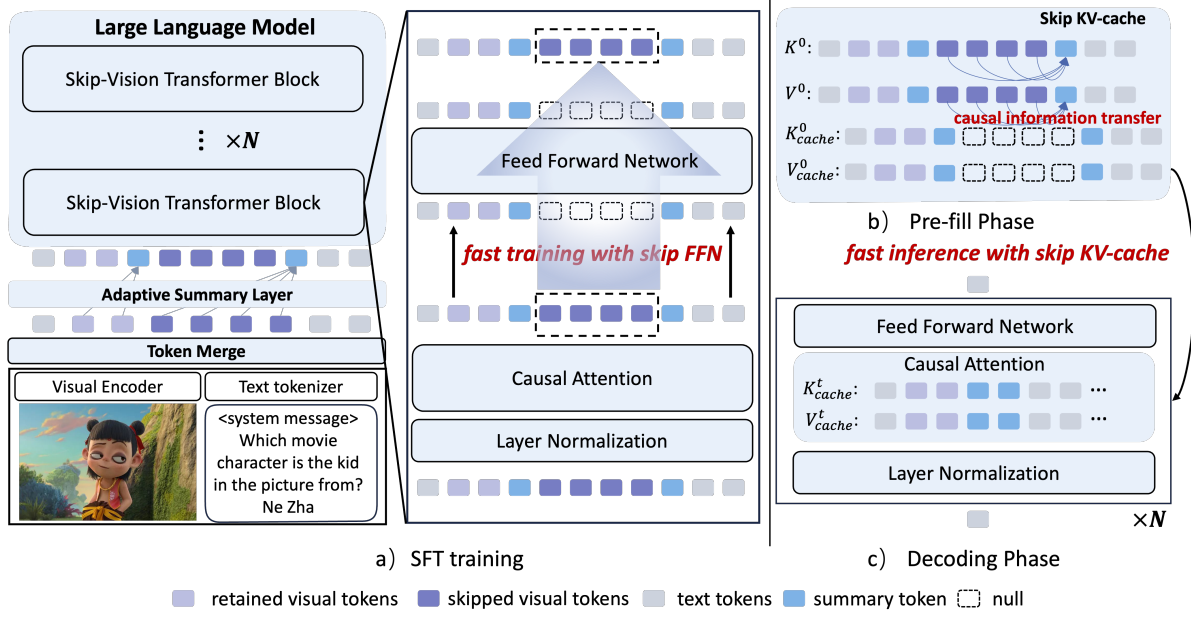


Figure 4. **The framework of Skip-Vision.** a) While visual scaling enriches visual information, it also increases computational overhead. Skip-Vision uses a skip-FFN strategy during training to reduce redundant computation for visual tokens. The numerous skipped tokens will be limited to the attention layer and bypass FFN. b) At the beginning of inference, Skip-Vision will remove skip-FFN visual tokens from the initial KV-cache, enhancing efficiency. c) During inference, skip attention leverages the skip KV-cache to accelerate generation.

where d_k is the dimension of K . This formulation allows each token to incorporate information from others based on relevance, making self-attention critical in capturing relationships across tokens.

Feed-Forward Neural Network: The feed-forward neural network enhances token representations following the self-attention layer. The operation can be expressed as:

$$\text{FFN}(x) = \text{Activation}(xW_1 + b_1)W_2 + b_2, \quad (2)$$

where x is the input, W_1 and W_2 are weight matrices, and b_1 and b_2 are biases. This architecture allows the model to capture complex patterns and improve the expressiveness of token representations.

KV-cache: The key-value cache (KV-cache) stores key (K) and value (V) representations from previous self-attention computations, enabling efficient reuse during inference. This approach avoids redundant calculations for earlier tokens, improving inference speed.

The attention score is computed using $\text{Attention}(Q, K_{\text{cache}}^t, V_{\text{cache}}^t)$, where Q is the query for the current token. At each time step t , the cache is updated as:

$$K_{\text{cache}}^{(t)} = \begin{bmatrix} K_{\text{cache}}^{(t-1)} \\ K_t \end{bmatrix}, \quad V_{\text{cache}}^{(t)} = \begin{bmatrix} V_{\text{cache}}^{(t-1)} \\ V_t \end{bmatrix}, \quad (3)$$

where K_t and V_t are the current token's key and value. This mechanism enhances computational efficiency and accelerates token generation.

4. Method

4.1. Skipped token selection and token merge

Before applying the skip FFN strategy, we first identify the skipped and retained visual tokens. The retained tokens should preserve most essential visual information, while the skipped tokens serve as complementary details, ensuring efficiency without significant loss of information.

Fortunately, in MLLM architectures, we can naturally distinguish retained and skipped tokens based on global and local context tokens. Global context tokens are retained, while local context tokens are skipped. In architectures like LLaVA-HD, global tokens originate from the resized full image, whereas local tokens come from smaller image patches. In CoS, global tokens correspond to larger window sizes, while local tokens are derived from smaller windows. Since LLaVA lacks a clear distinction between global and local tokens, inspired by [115], we select the top- n tokens most similar to the CLS token as retained tokens, categorizing the rest as skipped tokens.

To further reduce the computational burden, it is necessary to refine the visual tokens to eliminate excessive redundancy. As shown in Figure 5, we conducted a similarity analysis of the visual tokens in CoS at different levels, calculating their similarity density. We found that the smaller the granularity of the visual tokens, the higher their average similarity, while their token count is significantly greater than that of other granularities. Consequently, inspired by

ToMe [9], we employ a token merge method to reduce redundancy in skipped tokens, as detailed below:

- **Compute cosine similarities:** calculate the cosine similarity between each pair of tokens in the set, obtaining an average similarity score for each token.
- **Select top k distinctive Tokens:** sort tokens by average similarity in ascending order and retain the top k tokens with the lowest average similarity, representing the most distinctive information.
- **Merge remaining tokens:** for the remaining $n - k$ tokens, merge each one with the most similar token among the top k , thus consolidating information efficiently.

4.2. Training speed-up: skip FFN

As depicted in Figure 4a, we propose the FFN skip strategy: The retained visual tokens, which are few in number, proceed conventionally through all decoder layers of the LLM, while the numerous skipped visual tokens will be limited to the self-attention layers of each transformer block. For a given layer l , the output of retained tokens $h_{\text{retained}}^{(l)}$ is the sum of the self-attention output $h_{\text{attn}}^{(l)}$ and the FFN-transformed features $\text{FFN}^{(l)}(h_{\text{attn}}^{(l)})$:

$$h_{\text{retained}}^{(l)} = h_{\text{attn}}^{(l)} + \text{FFN}^{(l)}(h_{\text{attn}}^{(l)}). \quad (4)$$

The output of the skipped tokens is:

$$h_{\text{skipped}}^{(l)} = h_{\text{attn}}^{(l)}. \quad (5)$$

However, bypassing the FFN layers presents a non-trivial challenge. In the autoregressive training process, the final token of the vision sequence is used to predict the initial token of the text sequence, necessitating a specialized design for this terminal vision token. To facilitate this transition effectively, it must pass through the FFN layers, which are enriched with stored linguistic information [32, 83].

To address this, as shown in Figure 4, we introduce adaptive summary tokens to efficiently consolidate visual information. This token is generated by the Adaptive Summary Layer before the token sequence is input into the LLM. The layer comprises a simple linear transformation, and its computation is defined as follows:

$$x^s = (\text{softmax}(X \cdot W^T))^T \cdot X, \quad (6)$$

where X denotes the retained or skipped visual tokens, $x^s \in \mathbb{R}^{1 \times C}$ is the summary token, and $W \in \mathbb{R}^{1 \times C}$ is the weight matrix. Leveraging this layer, we concatenate a summary token on both ends of the skipped visual tokens before their input to the LLM. The first summary token aggregates essential information from retained tokens, mitigating potential loss from extended token sequences. The second summary token integrates prior sequence information and plays a key role in predicting the next text token.

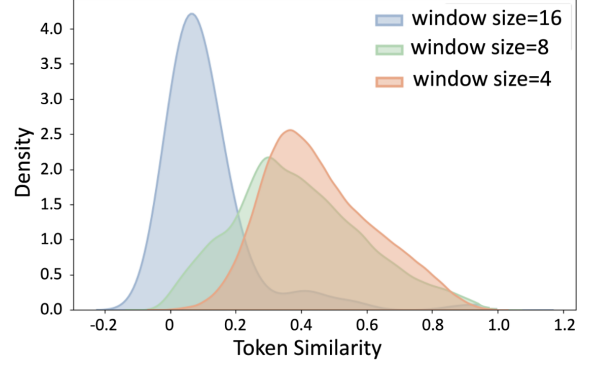


Figure 5. **Similarity density statistics.** We conducted a similarity density analysis in CoS, examining the visual tokens derived from various window scale. The results indicate that as the window scale decreases, the similarity between visual tokens becomes more pronounced, reflecting higher token redundancy.

4.3. Inference speed-up: skip KV-cache

After training, our Skip-Vision framework incorporates skip KV-cache method to further improve efficiency:

$$K_{\text{skip-cache}}^{(0)} = K_{\text{cache}}^{(0)} \setminus K_{\text{skip}}, \quad V_{\text{skip-cache}}^{(0)} = V_{\text{cache}}^{(0)} \setminus V_{\text{skip}}, \quad (7)$$

Equation 7 indicates that visual tokens skipped by FFN are removed from the KV cache after the pre-filling stage. As depicted in Figures 4 b) and c), given the autoregressive nature of LLMs, earlier token information is progressively transferred to later tokens. Skip FFN further enhances this integration by suppressing the attention weights of skipped tokens, forcing the model to focus on key tokens and improving information aggregation efficiency. Empirical results show that, due to the inclusion of summary tokens, this omission does not compromise model performance, underscoring a distinctive advantage of our framework for more efficient inference.

5. Theoretical analysis of Skip-Vision

At the core of this analysis lies the layer error incurred when bypassing the FFN layer. For a given layer l , the original output $h_{\text{original}}^{(l)}$ is : $h_{\text{original}}^{(l)} = h_{\text{attn}}^{(l)} + \text{FFN}^{(l)}(h_{\text{attn}}^{(l)})$. When skipping the FFN, the output becomes: $h_{\text{skip}}^{(l)} = h_{\text{attn}}^{(l)}$. The per-layer skipping error is:

$$\epsilon^{(l)} = \|h_{\text{original}}^{(l)} - h_{\text{skip}}^{(l)}\|_2 = \|\text{FFN}^{(l)}(h_{\text{attn}}^{(l)})\|_2. \quad (8)$$

For redundant tokens, such as those in homogeneous image regions, this error is negligible ($\epsilon^{(l)} \approx 0$) due to minimal feature transformations by the FFN.

However, errors propagate through subsequent layers, amplified by the recursive nature of transformer architectures. Leveraging Lipschitz continuity assumptions for self-

attention and FFN operations ($L_{\text{attn}}^{(l+1)}$ and $L_{\text{FFN}}^{(l+1)}$), the cumulative error at layer $l + 1$ is bounded by:

$$\epsilon^{(l+1)} \leq (L_{\text{attn}}^{(l+1)} + L_{\text{FFN}}^{(l+1)}) \cdot \epsilon^{(l)} + \epsilon_{\text{skip}}^{(l+1)}, \quad (9)$$

where ($\epsilon_{\text{skip}}^{(l+1)}$) represents new errors from skipping deeper layers ($l + 1$).

Over L layers, the total error telescopes to:

$$\epsilon_{\text{total}} \leq \sum_{l=1}^L \epsilon_{\text{skip}}^{(l)} \cdot \prod_{i=1}^{L-l} (L_{\text{attn}}^{(i+1)} + L_{\text{FFN}}^{(i+1)}). \quad (10)$$

Theorem 5.1 Bounded Lipschitz. *If the spectral norms of W_1, W_2, W_K, W_Q , and W_V are bounded by 1, then:*

- $L(\text{attn}) \leq 1/\sqrt{d_k}$
- $L(\text{FFN}) \leq 1$

Follow Theorem 5.1, we assume Lipschitz constants $L_{\text{attn}} + L_{\text{FFN}} \leq \gamma$ and skipping errors $\epsilon_{\text{skip}}^{(l)} \leq \epsilon$, the total error is scaled to:

$$\epsilon_{\text{total}} \leq \epsilon \cdot \frac{\gamma^L - 1}{\gamma - 1}, \quad (11)$$

Equation 11 establishes that the skip error is bounded when $\gamma < 1$, provided the model is trained with modern regularization techniques. This ensures that the MLLM remains less sensitive to the effects of skipping.

This error also impacts the KL divergence between the original and skipped outputs, bounded by:

$$\mathcal{D}_{\text{KL}}(p_{\text{skip}} \parallel p_{\text{original}}) \leq \frac{1}{2\sigma^2} \cdot \epsilon_{\text{total}}^2, \quad (12)$$

where σ^2 is the variance of the logits. Further integrating feature similarity errors ($\epsilon_{\text{sim}} = O(\sqrt{1-\theta})$) from low-attention tokens, the final bound becomes:

$$\mathcal{D}_{\text{KL}} \leq \frac{1}{2\sigma^2} \cdot (\epsilon_{\text{total}} + \epsilon_{\text{sim}})^2. \quad (13)$$

Practically, this analysis motivates a layer-wise skipping strategy, alongside token selection and merge based on feature similarity (θ). **More analysis and the proof of Theorem 5.1 are provided in the Appendix.**

6. Experiments

6.1. Benchmarks

To comprehensively evaluate our model, we conducted experiments on eight different benchmarks. These include MME [30], which assesses the perceptual and cognitive capabilities of multimodal language models; TextVQA [95], focused on fine-grained visual question answering; MM-Bench [72] and MMStar [21] for general diagnostic capabilities; MMMU [125], designed to test STEM-related reasoning; MathVista [76], which evaluates mathematical problem

solving skills; OCRBench [70], for optical character recognition tasks; and MMVet [121], used for subjective assessment. Taken together, these benchmarks offer a comprehensive framework for assessing the efficacy of the model in various multimodal challenges.

6.2. Efficiency evaluation

We systematically evaluate both training time (*hours*), inference FLOP computation and inference latency (*ms*) on single A100 GPU. For training efficiency, we report actual GPU hours measured on consistent hardware configurations. For inference FLOPs, we adopt the methodology from FastV [22], specifically tracking the computational demands of processing visual tokens. In the LLaMA architecture, we calculate FLOPs for the causal attention and feed forward network modules as $4NC^2 + 2N^2C + 3NCM$ where N is the token count, C denotes the hidden state dimension, and M is the intermediate FFN dimension. Skip-Vision framework adjusts token counts dynamically between the attention and FFN modules, allowing us to refine FLOP calculations as $4N_1C^2 + 2N_1^2C + 3N_2CM$, where N_1 and N_2 represent the visual token counts processed in the causal attention and FFN layers, respectively.

6.3. Data setup

During pretraining, we adopt the experimental settings of LLaVA and CoS, utilizing LLaVA-558K [69] and 65 million image-text pairs for respective comparisons. For the Supervised Fine-Tuning phase, we use LLaVA-665K [69] for all comparative experiments. By leveraging the computational efficiency of Skip-Vision framework, the resources saved can be redirected toward scaling the dataset to further enhance model performance. To this end, we extend our dataset to SV-1M, comprising 1 million samples, demonstrating the substantial performance improvements achievable through data expansion. To further assess our approach against state-of-the-art models of similar scale, we extend the dataset to SV-9M, underscoring the scalability and potential of our method. Detailed dataset statistics are provided in the Appendix.

6.4. Training setup

For LLaVA and LLaVA-HD setting, we use input resolutions of 336 and a dynamic resolution ranging from 336 to 672, respectively, with CLIP ViT-L/336px [86] as the vision encoder and LLaMA3 8B Instruct [29] as the backbone (additional comparisons under LLaVA-1.5-7B training setup are in the Appendix 10.1). For CoS setting, pretraining is conducted at 224 resolution, training only the multi-scale visual resampler with CLIP ViT-L/224px [86]. The model processes 80 visual tokens, including 16 global and 64 local tokens, and scales to 336 and 1296 tokens during SFT.

	MME	Textvqa	MMB	MMVet	MMM	MathV	OCRB	MMStar	Avg	Hours	FLOPs	Latency
LLaVA setting:												
LLaVA [69]	1535	58.6	72.2	32.8	39.6	21.3	32.7	36.5	42.0	10.2	100%	125
LLaVA-FastV [22]	1453	56.1	71	32.4	37.3	20.6	28.8	36.3	40.4	9.6	30%	103
MQT-LLaVA [39]	1487	53.6	69	28.2	37.8	16.8	28.1	36.5	38.6	7.5	44%	106
LLaVA-Tokenpacker [60]	1538	56.2	72.3	32.1	39.0	19.3	29.8	35.0	40.5	8.8	24%	97
SV-LLaVA	1519	56.5	72.5	32.4	38.8	21.2	30.7	36.5	41.3	7.9	25%	90
($N_r = 100, N_s = 156$)	-1%	-3.6%	+0.4%	-1.2%	-2%	-0.5%	-3.6%	-6.1%	-1.7%	-22.5%	-75%	-28%
SV-LLaVA	1530	57.3	72.7	35.6	39.2	21.4	32.0	40.0	42.6	9.0	60%	103
($N_r = 256, N_s = 320$)	-0.3%	-2.2%	+0.7%	+8.5%	-1%	+0.5%	-2.1%	+9.6%	+1.4%	-11.8%	-40%	-17.6%
LLaVA-HD setting:												
LLaVA-HD [67]	1533	64	71.8	35.3	39.3	20	37.4	40.5	44.0	19	100%	450
SV-LLaVA-HD	1534	61.2	73.1	35.9	39.7	21.2	34.1	39.3	43.5	12.2	50%	250
($N_r = 576, N_s = 576$)	+0%	-4.3%	+1.8%	+1.7%	+1.5%	-6%	-8.8%	-2.9%	-1.1%	-35.8%	-50%	-44.4%
CoS setting:												
CoS [40]	1585	64.4	77.1	39.4	39.2	21.5	39.2	41.2	46.0	15.3	100%	160
SV-CoS	1563	63.7	76.5	41.7	40.3	21.2	37	41.9	46.0	9.8	25%	90
($N_r = 272, N_s = 256$)	-1.4%	-1.1%	-0.8%	+5.8%	+2.8%	-1.4%	-5.6%	+1.7%	-0%	-40%	-75%	-43.8%
SV-CoS (SV-1M)	1569	67.1	77.1	43.3	40.7	22.0	50.7	44.4	49.3	16.3	25%	90

Table 2. **Performance and efficiency evaluation.** We evaluate Skip-Vision on LLaVA, LLaVA-HD, and CoS and compare it with state-of-the-art efficiency optimization models under the LLaVA setting. N_r and N_s denote the number of retained and skipped tokens, respectively.

	LLM	MME	Textvqa	MMB	MMVet	MMM	MathV	OCRB	MMStar
Open-source models in 8B tier:									
mPLUG-Owl3 [118]	Qwen2	-	69.0	<u>77.6</u>	40.1	-	65.0	-	40.1
Cambrian [100]	LLaMA3	1547	<u>71.7</u>	75.9	-	42.7	49.0	62.4	-
LLaVA [69]	LLaMA3	1535	58.6	72.2	32.8	39.6	21.3	32.7	36.5
Mini-Gemini-HD [61]	LLaMA3	1606	70.2	72.7	-	37.3	37.0	47.7	-
LLaVA-Next [56]	LLaMa3	<u>1604</u>	64.6	72.1	-	41.7	36.3	49.0	-
Ovis [77]	LLaMA3	-	-	77.4	-	<u>44.7</u>	40.8	-	<u>49.5</u>
MiniCPM-V2.5 [117]	LLaMA3	-	76.6	77.6	-	45.8	54.3	72.5	-
SV-CoS (SV-9M)	LLaMA3	1550	69.1	78.4	51.7	43	<u>61.3</u>	<u>69.4</u>	52.3

Table 3. **MLLM evaluation.** We present the performance of SV-CoS on SV-9M, comparing it against the current SOTA models of a similar scale. The highest scores are highlighted in **bold**, while the second-highest scores are indicated with underlines.

In SFT, the resolution increases to 448, and all model parameters are unfrozen for full optimization.

In all experiments, we used a learning rate of $1e^{-3}$ for pre-training, which is reduced to $1e^{-5}$ during SFT. Training follows a cosine decay schedule after a 3% warm-up phase. For skipped token selection, unless otherwise specified, the LLaVA setting includes 156 skipped tokens and 100 retained tokens, the LLaVA-HD setting has 576 skipped and 576 retained tokens, and the CoS setting consists of 256 skipped and 272 retained tokens. All skipped tokens undergo a token merging process to enhance efficiency.

6.5. Main experimental results

Performance and efficiency. We first evaluate the effectiveness of Skip-Vision across three different frameworks: CoS, LLaVA, and LLaVA-HD, demonstrating its ability to maximize efficiency while minimizing performance degradation. As shown in Figure 1 and Table 6, Skip-Vision re-

duces training time by 22.5%, inference FLOPs by 75%, and inference latency by 28% for LLaVA. For LLaVA-HD, it achieves a 35.8% reduction in training time, 50% in inference FLOPs, and 45% in inference latency. In CoS, it reduces training time by 35.8%, inference FLOPs by 75%, and inference latency by 44%.

Scaling. We evaluated the scalability of Skip-Vision, leveraging CoS’s visual scaling to increase visual tokens from 336 to 1296 while maintaining efficiency. Expanding the fine-tuning dataset from LLaVA-665k to SV-1M further improved performance by 8% over CoS_{1296} with similar resources. To benchmark against state-of-the-art models, we scaled to SV-9M, with training completed in 72 hours on 16 NVIDIA A100 GPUs (Table 3).

6.6. Ablation study and analysis

To validate the effectiveness of each component within the Skip-Vision framework, we conducted ablation and com-

	MME	Textvqa	MMB	MMVet	MMMU	MathV	OCRB	MMStar	Avg	Latency
SV-LLaVA (w/o SK)	1519	56.7	72.5	31.4	38.8	21.2	30.9	36.5	41.1	103
SV-LLaVA	1519	56.5	72.5	32.4	38.8	21.2	30.7	36.5	41.3	90
	+0%	-0.4%	+0%	+3.2%	+0%	+0%	-0.6%	+0%	+0.5%	-9.7%
SV-LLaVA-HD (w/o SK)	1533	61.6	73.0	35.8	39.9	21.2	34.5	39.7	43.7	310
SV-LLaVA-HD	1534	61.2	73.1	35.9	39.7	21.2	34.1	39.3	43.5	250
	+0%	-0.7%	+0.1%	+0.3%	-0.5%	+0%	-1.2%	-1%	-0.5%	-19.4%
SV-LLaVA-CoS (w/o SK)	1562	63.9	76.5	40.2	40.2	21.2	37.2	41.9	45.9	126
SV-LLaVA-CoS	1563	63.7	76.5	41.7	40.3	21.2	37	41.9	46.0	90
	+0%	-0.3%	+0%	+3.7%	+0.2%	+0%	-0.5%	+0%	+0.2%	-28.6%

Table 4. **Ablation on skip KV-cache.** We evaluate SK (using skip KV-cache during inference) of Skip-Vision on different settings.

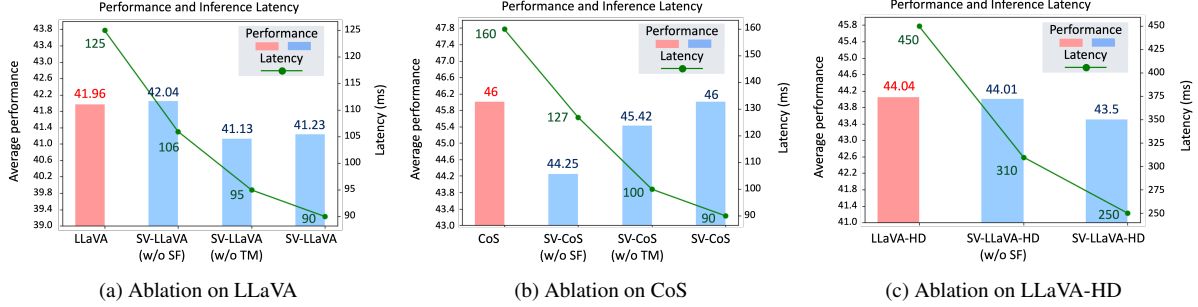


Figure 6. **Ablation study.** We performed an ablation study on each component of the Skip-Vision. SF (skip FFN), TM (token merge).

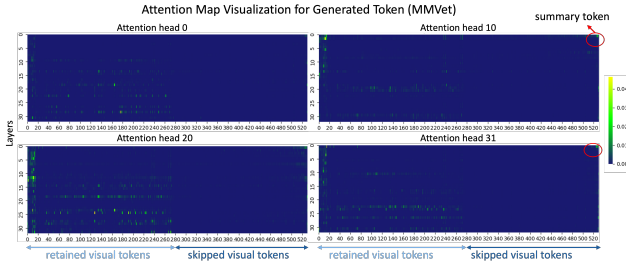


Figure 7. **Visualization of attention map in MMVet.**

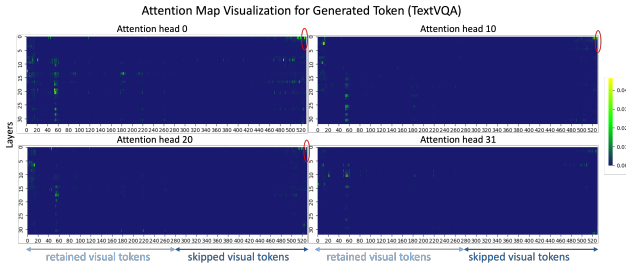


Figure 8. **Visualization of attention map in TextVQA.**

parative experiments on token merge, skip FFN, and skip KV-cache.

As shown in Figure 6, both the skip FFN strategy and token merging enhance computational efficiency across different model settings while maintaining performance. Notably, skip FFN performs better in the CoS setting than in LLaVA and LLaVA-HD. We attribute this to CoS allowing the vision backbone to be fine-tuned during the SFT stage, enabling key information to be more effectively captured by

summary tokens, thereby reducing the errors introduced by skip FFN. As shown in Table 4, skip KV-cache slightly affects fine-grained tasks such as TextVQA and OCRBench but improves performance in long-form response tasks like MMVet.

To further analyze this, we visualized attention maps across layers for visual tokens in SV-CoS during the MMVet and TextVQA tasks. In MMVet (Figure 7), the attention of generated tokens primarily focuses on retained and summary tokens, with skip KV-cache filtering out unnecessary noise. In TextVQA (Figure 8), which requires fine-grained understanding, attention is spread across both retained and skipped tokens, leading to some performance loss with skip KV-cache. However, causal attention effectively aggregates information from skipped tokens into the final summary tokens, allowing us to retain only the summary tokens and retained visual tokens in the KV-cache.

7. Conclusion

We propose Skip-Vision, an efficient multimodal large language model architecture that boosts computational efficiency through a skip-FFN strategy, reducing redundant computations over visual tokens during training. During inference, Skip-Vision uses a skip KV-cache to accelerate processing by omitting non-essential visual tokens from the KV cache. Extensive experiments show our model outperforms current methods in efficiency and scales effectively with data, delivering competitive performance among leading models of similar scale.

8. Acknowledgements

This work was supported in part by NSFC (62201342), and Shanghai Municipal Science and Technology Major Project (2021SHZDZX0102). This work was supported by Ant Group Research Intern Program.

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